Words → boundary & initial conditions

1 · Method

1. **PDE:** temperature → heat; displacement → wave; "steady / equilibrium" → Laplace; "infinitely long" → FT.
2. **List every boundary first:** each end / edge, $r = 0$, θ edges, $\pm\infty$.
3. **One sentence per boundary.** Fixes the *value* → Dirichlet. Fixes the *slope / heat flow* → Neumann. A *relation* → Robin.
4. **Unmatched boundary → hidden condition:** bounded, periodic, decay.
5. **Time:** "initially" → IC. "For $t > 0$ / suddenly" → BC. "After a long time" → $u_t = 0$.

2 · Count the conditions

Problem	Needs
1D heat	2 BCs + 1 IC
1D wave	2 BCs + **2 ICs**
Laplace, rectangle	4 BCs (one per edge)
Laplace, full disc	$r = a$ + **bounded at $r = 0$** + **periodic**
Laplace, annulus	$r = a$, $r = b$ + **periodic**
Semicircle / wedge	2 straight edges + arc + **bounded at $r = 0$**
Semi-annulus	4 edges, nothing hidden
Axisymmetric disc	$r = a$ + **bounded at $r = 0$**, no θ
Infinite line (FT)	$u, u_x \to 0$ **as** $\lvert x\rvert \to \infty$

3 · Heat: phrase → BC

Words	Condition
held / kept at T_0, bath, reservoir	$u = T_0$
ice (0), boiling water (100)	$u = 0$, $u = 100$
insulated, no heat flow, adiabatic, reflecting	$u_x = 0$
temperature gradient c	$u_x = c$
heat *in* at rate q	$u_x(0) = -\frac{q}{K}$, $u_x(L) = \frac{q}{K}$
heat *out* at rate q	signs flip
Newton cooling into air at T_s	$u_x(L) = -\beta(u - T_s)$
	$u_x(0) = +\beta(u - T_s)$
end temperature set to $g(t)$	$u(0,t) = g(t)$
wire bent into a closed loop	periodic

Sign rule: $K\dfrac{\partial u}{\partial n}$ = heat flux *into* the body (n outward: $+x$ at L, $-x$ at 0). Check: heat flows hot → cold. No K given ⇒ $K = 1$.

Not BCs (change the PDE):
"insulated except at its ends", "laterally insulated", "faces insulated" → just 1D / 2D.
"heat generated" q or $\propto x$ → $+q$ or $+cx$ in the PDE.
"lateral surface loses heat" → $-\beta(u - T_s)$ in the PDE.

4 · Heat ICs

"initially $f(x)$" → $u(x,0) = f$. "Uniform T_1" / "from an oven at T_1" → $u(x,0) = T_1$.
"Steady with ends at A, B for a long time; at $t = 0$ …" → IC = *old* steady state $A + \frac{(B-A)x}{L}$, BCs = *new* ones.
IC and BC may disagree at a corner: fine.

5 · Wave

Words	Condition
fixed, pinned, clamped, tied	$u = 0$
free end, ring on frictionless rod	$u_x = 0$
end shaken as $h(t)$	$u = h(t)$
released / let go / from rest	$u_t(x,0) = 0$
rest / equilibrium position, flat	$u(x,0) = 0$
pulled into shape f	$u(x,0) = f(x)$
struck, given velocity g	$u_t(x,0) = g(x)$

Plucked at $x = \ell$ to height A, from rest:

$$u(x,0) = \begin{cases} \dfrac{Ax}{\ell}, & 0 \le x \le \ell \\ \dfrac{A(L-x)}{L-\ell}, & \ell \le x \le L \end{cases}$$

6 · Laplace, polar, hidden

Rectangle: insulated left / right → $u_x = 0$; top / bottom → $u_y = 0$.
Polar: insulated arc → $u_r = 0$; insulated straight edge → $u_\theta = 0$.
Semicircle: "diameter" = edges $\theta = 0$ and $\theta = \pi$.
Bounded at $r = 0$ (centre in domain): kills $\ln r$, r^{-n}, Y_0.
Periodic (θ fully round):

$$u(r,-\pi) = u(r,\pi)$$
$$u_\theta(r,-\pi) = u_\theta(r,\pi)$$

Exterior $r > a$: bounded as $r \to \infty$.

7 · "After a long time"

Set $u_t = 0$; IC not used, except:
both ends insulated → $u \to \frac{1}{L}\int_0^L f\,dx$ (mean of IC);
both Neumann, net flux in $\neq 0$ → no steady state;
all-Neumann Laplace → unique only up to a constant.

8 · Then

Homogeneous BCs transfer ($T \not\equiv 0$):

$$u(a,t) = 0 \;\Rightarrow\; X(a) = 0$$
$$u_x(a,t) = 0 \;\Rightarrow\; X'(a) = 0$$

Inhomogeneous → $u = u_S + v$; v gets the *same type* with RHS 0, IC $f - u_S$.

BCs on $[0, L]$	Eigenfunctions
D–D	$\sin\frac{n\pi x}{L}$, $n \ge 1$
N–N	1, $\cos\frac{n\pi x}{L}$, $n \ge 0$
D–N	$\sin\frac{(2n-1)\pi x}{2L}$, $n \ge 1$
N–D	$\cos\frac{(2n-1)\pi x}{2L}$, $n \ge 1$
periodic $[-L, L]$	1, $\cos\frac{n\pi x}{L}$, $\sin\frac{n\pi x}{L}$

9 · Traps

1. "Insulated *except at its ends*" describes the sides, not the ends.
2. Fixed string → $u = 0$. Fixed *temperature* T_0 → $u = T_0$.
3. "From rest" = velocity 0. "Rest position" = displacement 0.
4. Wave always needs two ICs; find the hidden one.
5. Flux and Newton signs flip between $x = 0$ and $x = L$.
6. Sources and lateral loss go in the PDE, not the BCs.
7. Disc: bounded + periodic. Annulus: periodic only. Wedge: bounded only.
8. Check which end is $x = 0$ before using the eigen-table.

10 · Write it like this

$$\begin{aligned} &u_t = \alpha u_{xx}, && 0 < x < L,\ t > 0 \\ &u(0,t) = \ldots, && t > 0 \quad \text{"in ice"} \\ &u_x(L,t) = \ldots, && t > 0 \quad \text{"insulated"} \\ &u(x,0) = f(x), && 0 < x < L \end{aligned}$$

Domains on every line. Then say: homogeneous or not.